

MARKETING ANALYSIS REPORT

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SUMMARY:

This report presents the findings and recommendations from the recent CodeWay marketing analysis, covering three key areas:

- Debugging (Q1): Diagnosed and resolved an anomalous spike in the Purchaser User Rate metric on July 9, 2024.
- Dashboard (Q2): Developed a comprehensive Looker Studio dashboard to monitor daily app performance and user behavior for the X application.
- Marketing (Q3): Attributed revenue to acquisition channels and comparison of the performances of Facebook versus Apple Search Ads.

The files and codes related to this analysis are included in the file containing this report.

Q1: Debugging

- a) Investigation of the unexpected spike on 9th of July was carried out as follows:
 - Calculated daily unique active users.
 - Examined 9th of July purchase records and unique purchaser counts.
 - There was a total of 1075 purchase counts while 796 unique purchasers.
 - Further investigation indicated that there were 279 duplicate entries which each pair has the same user id, event time, and iOS operating system.
 - After removing the duplicates, Purchaser User Rate dropped to 1.88% which is a more expected percentage.

The spike was probably caused by an iOS logging bug, and it can be said that this data provided wrong information.

- b) There were 42368 active users, and 796 unique purchasers recorded on 9th of July. Thus, 0.66% of the active users, and 35% of the unique purchasers were affected. All affected users were on iOS, suggesting the issue may have been platform specific.

Q2: Dashboard

The dashboard was designed using Looker Studio. Before importing the provided data, a .csv file was created which includes computed daily metrics, and split event names columns. This preprocessing step enabled easier Looker Studio in-built calculations and creation of fields. A PDF version is included in the case study file and the corresponding Looker Studio report link is as follows:

<https://lookerstudio.google.com/reporting/3fc8d609-e73b-43bf-b3c5-704f8d20a68d>

Q3: Marketing

- a) The investigation was conducted using Google BigQuery. Steps included:
- Loaded marketing_events and marketing_users files into BigQuery tables.
 - Exploded the tracker_names array to match each user to single ads Platform.
 - Exploded the properties array to extract revenue values.

The resulting table (3.1) is formed as desired.

Platform	Revenue (USD)
Apple	81.14
Facebook	102.82
Organic	293.13

Table 3.1: Revenue Table

- b) The comparison of Facebook and Apple Search Ads was carried out as follows:
- Counted the unique installs per platform.
 - Calculated the total ad spend, return on ad spend (ROAS), and revenue per install.

The table below (3.2) shows the results of the calculations.

Platform	Installs	CPI(USD)	Total Cost (USD)	Total Revenue (USD)	ROAS	Rev/Install (USD)
Facebook	48	1.30	62.40	102.82	1.65	2.14
Apple	26	3.10	80.60	81.14	1.01	3.12

Table 3.2: Facebook vs. Apple

The calculations indicate that Facebook ads have higher return on ad spend, higher positive surplus while Apple ads grant higher revenue per install and just pays its price. To sum up, both platforms are effective, but Facebook platform is more efficient and has better overall performance.